SELVA KARUPPASAMY M

AI Backend Engineer

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SelvaKaruppasamyM

ABOUT ME

Backend Engineer with over 4 years of experience specializing in Python Tooling and high-performance System Architecture at Zentron Labs. Proven track record of bridging product vision with technical execution through cross-functional collaboration, delivering scalable, enterprise-grade solutions. Skilled in modern AI-driven development methodologies (TDD/SDD) and focused on driving measurable business value through optimized backend infrastructure.

WORK EXPERIENCE

Associate Software Developer 07/2022 – Present

Zentron Labs Pvt Ltd

- Collaborated with Product Owners to define technical requirements and delivered interactive Figma prototypes, accelerating stakeholder alignment by validating workflows prior to development and reducing initial feature deployment time by an estimated 1 week.
- Architected a scalable Python backend for a Computer Vision-as-a-Service (CVaaS) platform utilizing FastAPI and SQLAlchemy, capable of supporting 50 concurrent asynchronous tasks and handling 100+ MB of image data.
- Modernized legacy compute functions by implementing optimized Python modules via CFFI, reducing image processing latency by 50% and increasing overall system throughput to 15 transactions/frames per second."

SKILLS

Languages & Frameworks
Python (FastAPI, Flask, SQLAlchemy),SQLite,PostgreSQL,CFFI.

Al & Automation
OpenCode Agent, Playwright MCP, TDD, SDD.

System & Design
System Design, Microservices, Figma, REST API, Git, SVN, CI/CD Pipelines, WSL.

Vision & Device control
LabVIEW, NI-VISA, NI-IMAQdx.

PROJECTS

Agentic LabVIEW to Python Compiler
Architected a multi-stage LLM pipeline to automatically translate visual LabVIEW code into Python/C++. Built a decoupled architecture featuring a CustomTkinter GUI frontend and specialized backend agents for parsing XML logic, processing JSON data flows, and analyzing block diagram images using the Gemini API.

Real-Time Part Configurator
Developed a responsive, real-time configurator platform from the ground up using Elixir and Phoenix LiveView. Applied SDD/TDD methodologies alongside AI backend tooling to build seamless, interactive user experiences with minimal latency..Integrated Playwright MCP to facilitate advanced AI-driven development and testing workflows.

CVaaS Infrastructure Development
Built a high-performance Python backend featuring asynchronous processing, PostgreSQL database integration, and real-time health monitoring for computer vision tasks.

Industrial Configurator Architecture
Designed a modular framework for an industrial configurator integrating C++ DLLs, reducing new feature integration time from 1 week to 3 days while minimizing technical debt.

EDUCATION

Bachelor of Electrical and Electronics Engineering 07/2018 – 04/2022
Alagappa Chettiar Government College of Engineering and Technology TamilNadu, India
CGPA: 8.6/10